train the model on various values of $\alpha \in \{10^{-5}, 10^{5}\}$, and then choose the α that yields the highest correlation. The training time per timestep in both 2D and 3D is less than 10 minutes on a standard CPU.

F. GlassMLP

The methodology employed in this manuscript corresponds exactly to the procedure described in detail in Ref. [S5], including all training meta data. In particular, we calculate a set of M_S physics-inspired descriptors, including the coarse-grained local density, $\overline{\rho}^i_{L,\beta} = \sum_{j\in N^i_\beta} e^{-r_{ij}/L}$, which is coarse-grained over a distance L by summing over all N^i_β particles of type β within distance $r_{ij} = |\boldsymbol{r}^{\rm inh}_i - \boldsymbol{r}^{\rm inh}_j| < 20\sigma_{11}$ of particle i. Importantly, particle positions are extracted from the inherent structures $\boldsymbol{r}^{\rm inh}_i$, and not directly from the thermal structure. Similarly, we include the potential energy, perimeter of the Voronoi cell and local variance of potential energy (see Ref. [S5] for details).

Different from Ref. [S5] we additionally include one further structural descriptor, based on the distance between the inherent and the thermal states of particle i, $\Delta r^{\rm inh}_i = |\boldsymbol{r}^{\rm inh}_i - \boldsymbol{r}^{\rm th}_i|$: $\overline{\Delta r}^i_{L,\beta} = \sum_{j\in N^i_\beta} \Delta r^{\rm inh}_i e^{-r_{ij}/L}$. This descriptor does not affect the long-time prediction and was therefore discarded in Ref. [S5], but is reintroduced here mainly to improve the short-time performance of GlassMLP.

To predict propensity in the KA (3D) system, GlassMLP has in total 618 free parameters. In the ternary KA2D system, the number is slightly increased (765). The training just takes 2-5 minutes with an octa-core CPU or a Laptop GPU (NVIDIA T600 Laptop).

G. SE(3)

The model is exactly that presented as the main model of Ref. [S11] (version 2, august 2023).

We recall the main features for completeness and to avoid possible confusion. In the SE(3) GNN we use 8 layers. The structure of each layer is the same: 8 channels for each l component, with $l = 0, 1, 2, 3$ (namely, $l_{max} = 3$). The input data consists of the thermal positions (not quenched to the IS) and of the particles potential energy (but computed from the inherent state's particles positions). We predict simultaneously propensities for all particle types and time steps. The connectivity graph is computed using a threshold distance of $2.0\sigma_{11}$. The input node features are the particle type (one-hot encoded) concatenated with the potential energy (of the IS). For the 2D case, we introduce an artificial third component to the position of all particles ($\forall i, z_i = 0$) and used spherical harmonics embedding. This is overkill but is a quick way to adapt the 3D scheme to 2D data.

The radial-encoding MLP encodes the radius on (Bessel) basis functions using 10 basis vectors, followed by a hidden layer of 16 neurons, and uses a dropout of 0.3. We use batch normalization between each convolution layer. We minimise the MSE loss with a L^2 regularization coefficient $\beta = 10^{-7}$, using a batch size of 2 graphs (8 in 2D), a learning rate of 10^{-3} with Adam ($\beta_1 = 0.99$). As opposed to Ref. [S11], we do not use a validation set in this manuscript and therefore do not perform early stopping. We simply take the last epoch model (the accuracy and losses are basically flat when we interrupt learning).

The model has $52,660$ parameters for 3D data (respectively $53,394$ in 2D). We perform 200 epochs (100 in 2D), which takes approximately 11 hours (27 hours in 2D) using an NVIDIA Tesla V100 (32GB) (the CPUs on the node are 2 Intel Xeon Gold 6148 20 cores (40 threads) at 2.4 GHz (Skylake)). Note that we train simultaneously the 10 time steps (6 for KA2D) of a given state point (temperature), therefore the training does not have to be repeated for each timescale.

H. SBO

Unsupervised learning of local structure fluctuations is carried out for the 3D KA samples along the lines of Ref. [S12], using the `partycls` Python package [S13]. Namely, we characterise the local structure around particle i using the smooth bond-order (SBO) descriptor

$$X^{\rm SBO}(i) = (Q^S_0(i), \ldots, Q^S_{l_{\rm max}}(i)), \tag{S2}$$

where Q^S_l are smoothed bond-orientational invariants of order l computed over the first coordination shell of particle i, see Refs. [S12] for full details. The maximum order $l_{\rm max}$ is equal to 8. A PCA is then carried out for each chemical species, to identify the directions in the descriptor space that capture the largest structural fluctuations. In particular, the projection on the first principal component, $\tilde{X}^{\rm SBO}_1(i)$, is a measure of structural heterogeneity that is correlated to some extent with the dynamics of glassy binary mixtures [S12]. The correlation between the first projection of the SBO descriptor and the propensity of motion is calculated directly on the test dataset, since there is usually no separate training stage in unsupervised learning. Finally, along with the bare SBO descriptor, we consider a coarse-grained (CG) version

$$\tilde{X}^{CG}_i = \frac{\sum_j \tilde{X}_j \cdot w(r_{ij}; L)}{\sum_{j=1}^N w(r_{ij}; L)}.$$

where the sum over j runs over the particles of the same species as i. As in Ref. [S14] we use an exponential function $w(r; L) = e^{-r/L}$, and we set $L = 1$.

I. Other structural descriptors

In addition to these models we have also included several traditional structural descriptors.

Density:
The density is connected to the structural input of GlassMLP (see definition Sec. I F above). In particular, we have used $\overline{\rho}^i_{L=5,\beta=\text{all}}$, i.e. coarse-grained over a distance $L = 5$ considering particles of all types. The only difference to GlassMLP is that we have evaluated the density in the thermal states.

Potential energy $\overline{E}^{\text{inh}}_{\text{pot}}$:
The potential energy in the inherent state, $\overline{E}^{\text{inh}}_{\text{pot}} = \overline{E}^i_{L=5,\beta=\text{all}} = \sum_{j\in N^i_\beta} E^j e^{-r_{ij}/L}/\overline{\rho}^i_{L,\beta}$, is similarly part of the structural descriptors which is used as input for GlassMLP [S5]. Here, $E^i = \sum_{j\neq i} V(r_{ij})/2$ is the potential energy of particle i.

Bond-order 3D (Ψ_6):
The bond-order descriptor Ψ^i_6 of particle i for the KA system in three dimensions is defined via the complex coefficient [S15]

$$q^i_{l,m} = \frac{1}{N_b}\sum_{j\in N_b} Y^m_l(\boldsymbol{r}_{ij}), \tag{S3}$$

where N_b is the number of bonds of particle i, defined as all neighbours within a cutoff of $r_{\text{cut}} = 2.0\sigma_{11}$, and $Y^m_l(\boldsymbol{r}_{ij})$ are the spherical harmonics. Using q^i we define,

$$\Psi^i_l = \sqrt{\frac{4\pi}{2l+1}\sum_{m=-l}^{m=l}\left|q^i_{l,m}\right|^2}. \tag{S4}$$

As before, we also further coarse-grain the descriptor, $\overline{\Psi}^i_{6,L=5,\beta=\text{all}} = \sum_{j\in N^i_\beta} \Psi^j_6 e^{-r_{ij}/L}/\overline{\rho}^i_{L,\beta}$.

Bond-order 2D (Ψ_4):
For the KA2D system, we slightly adapt the definition,

$$q^i_l = \frac{1}{N_b}\sum_{j\in N_b} e^{\mathrm{i}l\theta_{ij}}, \tag{S5}$$

which is strongly related to the n-fold symmetry defined in Sec. I E. From this we calculate,

$$\Psi^i_l = \sqrt{q^i_l\cdot(q^i_l)^*}, \tag{S6}$$

which is subsequently coarse-grained in the same way as defined in the 3D case.

Tanaka's Θ order parameter:
In Ref. [S16] Tong and Tanaka proposed the usage of a structural descriptor Θ^i, which quantifies the strength of local packing around particle i (see Ref. [S16] for details and definitions). We similarly coarse-grain this descriptor $\overline{\Theta}^i_{L=5,\beta=\text{all}} = \sum_{j\in N^i_\beta} \Theta^j e^{-r_{ij}/L}/\overline{\rho}^i_{L,\beta}$.

In a 2D system of polydisperse harmonic spheres, Ref. [S16] reports a Spearman's rank correlation coefficient of around $\rho_{\text{S}} \approx 0.9$ for the coarse-grained $\overline{\Theta}$ with propensity. ρ_{S} is usually strongly related to Pearson correlation ρ_{P}. In this roadmap, we show that the performance is significant reduced when applied to systems which are not prone to crystallization. Additionally, some of the performance difference might emerge from calculating Pearson correlation over all particles independent of their radii in Ref. [S16], as opposed to making independent predictions for each particle type as done in this manuscript. This emphasises the importance of using standardized datasets and performance metrics to quantify and validate the performance of newly developed descriptors and techniques.

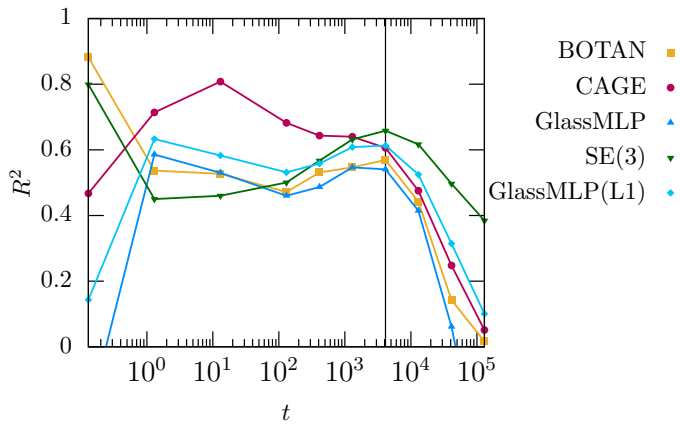

FIG. S1. Coefficient of determination $R^2 = 1 - \sum_i \left(\mathcal{R}^{\text{MD}}_i - \mathcal{X}^{\text{ML}}_i\right)^2/\text{var}(\mathcal{R}^{\text{MD}}_i)$ between the model predictions and the ground truth in the KA system (3D) at $T = 0.44$ for various ML models. The dynamical variable is the propensity of displacements. The vertical line marks the structural relaxation time τ_α. GlassMLP(L1) denotes the GlassMLP model [S5] trained without the additional terms in the loss function using only the L^1 norm.

II. ADDITIONAL BENCHMARKING

In the following, we will discuss additional metrics and ways to visualise the performance of the various methodologies which have been benchmarked intensively in Sec. V of the main text.

A. Coefficient of determination R^2

The coefficient of determination $R^2 = 1 - \sum_i \left(\mathcal{R}^{\text{MD}}_i - \mathcal{X}^{\text{ML}}_i\right)^2/\text{var}(\mathcal{R}^{\text{MD}}_i)$ is a popular measure to quantify the quality of a fit, including, in particular, supervised ML models. In Fig. S1 the coefficient of de-

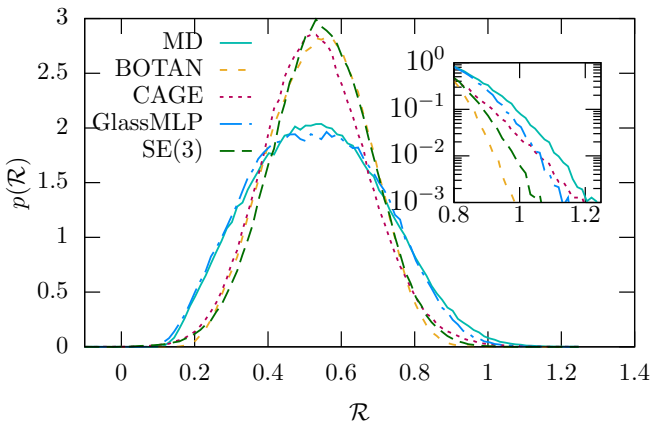

FIG. S2. Probability distribution of predicted and simulated propensities at $t = \tau_\alpha$ for the KA system (3D). The dynamical variable is the propensity of displacements $\mathcal{R}$. The inset shows a zoom to the tail, with logarithmic y-axis.

termination is shown for exactly the same data as was analysed with the Pearson correlation in Fig. 3. It can be observed that, in general, the two measures ρ_P and R^2 do not show very different results. In particular, SE(3) still features the best performance at $t = \tau_\alpha$. The most pronounced difference is that GlassMLP performs worse when measured with R^2 instead of ρ_P. This can be explained by the usage of a loss function which contains additional terms on top of the standard L^1 or L^2 norm [S5]. To validate this explanation we show the results for GlassMLP(L1) trained without these additional contributions and find results significantly better than for the original GlassMLP (see Fig. S1).

B. Probability distribution of propensities

Another interesting observable is the probability distribution of the propensities of displacement, $p(\mathcal{R})$, which can be extracted both from the MD simulations and from the ML models. A similar analysis has been performed in Ref. [S5]. As can be seen in Fig. S2 the propensities predicted by the models tend to underestimate the variance of the distribution. This observation can be rationalized, since predicting outliers could be very costly in the loss function during training, hence networks tend to predict small variances. The only network which agrees quite well to the MD prediction is GlassMLP, which has been constructed to adapt the variance by an explicit contribution in its loss function [S5]. However, as can be seen in the inset of Fig. S2, even for GlassMLP the tail of strongly moving particles is not perfectly reproduced. Focusing on this tail would likely significantly worsen the overall performance of the model.

C. Scatter plots of propensity predictions

The Pearson correlation ρ_P or the coefficient of determination R^2 are very good measures to quantify the performance of a network using a scalar quantity (see Figs. 3-6 in the main text and Fig. S1). However, they obviously cannot capture all details of the connection between the true and predicted propensities. In Fig. S3 we therefore show a 2D histogram visualizing this connection in greater detail. The overall impression of these figures is very similar to what we have concluded before. The most obvious difference between the methods is that GlassMLP is symmetric around the straight line, while the other techniques are slightly asymmetric. This observation is connected to the differences in the probability distribution as discussed in the previous paragraph.

D. Snapshots

The best way to visualise the predictions are snapshots of the amorphous structure, where each particle is coloured according to its propensity $\mathcal{R}$. In Fig. S4 we show an exemplary structure and the predicted propensities of displacement for various different models. Generally, strong correlations between the models and the MD simulations can be observed, as was expected from the high Pearson correlations. The strongest visible difference between MD and the ML models is that the latter are much smoother, in particular the ones based on GNNs (BOTAN and SE(3)).

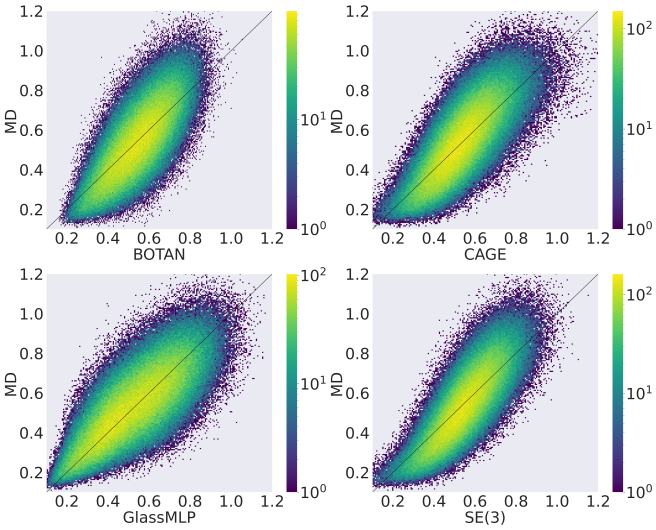

FIG. S3. 2D histograms of particle propensity $\mathcal{R}_i$ to compare different ML techniques to the MD ground truth in the KA system (3D, $t = \tau_\alpha$). Each entry in the histogram represents one particle, i, where the x/y-coordinates correspond to the propensities as predicted by ML and from MD simulations, respectively.